Problem FoL.1 ... misbehaved students

On a planet far, far away, some misbehaved students threw an object out of a window, locate
$h = 40\,\mathrm{m}$ *above the ground. It landed after* $t = 4\,\mathrm{s}$. *How much would this object weigh on Earth*
if the gravitational force acting upon the object on the distant planet is $F = 55\,\mathrm{N}$? *Conside*
the gravitational field to be constant on both planets and the air drag to be negligible.

Olda was having plenty of time while Náry was cooking

Problem FoL.2 ... plastered block

Kuba is boasting that he can lay a perfectly smooth cuboid on an inclined plank in such a wa
hat the cuboid won't slip down. The plank is inclined at an angle $\alpha = 20^\circ$ to the horizonta
plane. Find out what Kuba's acceleration has to be when he holds the plank, so that the cuboi
would remain at rest (with respect to the plank). *Mirek doesn't believe such silly tricks. .*

Problem FoL.3 ... we're on a roll

A truncated cone is rolling on a table in such a way, that a point on the perimeter of its smalle base moves at a speed $v_1 = 1.5\,\mathrm{m{\cdot}s^{-1}}$ *with respect to the base's center and a point on th perimeter of its greater base moves at a speed* $v_2 = 1\,\mathrm{m{\cdot}s^{-1}}$ *(again, with respect to the base' center). The slant height is* $l = 0.1\,\mathrm{m}$*. How long will it take the cone to return to the point fro which it started rolling?*

Tom dropped a pestle

Problem FoL.4 ... cooling

Consider a Peltier cell with cooling power $P = 10\,\mathrm{W}$*, placed on a person's wrist. Let's assum*
that all of its power is spent on cooling blood flowing through a vein with volumetric flo
rate $Q = 1.6\,\mathrm{ml{\cdot}s^{-1}}$*. Blood density is* $\varrho = 1\,025\,\mathrm{kg{\cdot}m^{-3}}$*, its heat capacity* $c = 4.2\,\mathrm{kJ{\cdot}kg^{-1}{\cdot}K^{-1}}$
How much cooler is the blood after flowing through the wrist?

Filip had to improvise in the summer

Problem FoL.5 ... wet road

How much does the shortest braking distance increase when it rains, if we do not want the ca
o skid? The car is moving at a speed of $v = 90\,\mathrm{km{\cdot}h^{-1}}$*, the coefficient of friction between a dr*
oad and the tires is $f_1 = 0.55$ *and it changes to* $f_2 = 0.30$ *in wet conditions. The weight o*
he car is $m = 1\,500\,\mathrm{kg}$ *and the acceleration due to gravity is* $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$*. Consider onl*
inetic energy of the car's translational motion. *Pikoš barely managed to stop*

Problem FoL.6 ... high five!

Tomáš and Michal have met on a cycleway and want to high five while riding. Both of them have the same mass $m = 70\,\mathrm{kg}$, *speed* $v = 8\,\mathrm{m{\cdot}s^{-1}}$ *and their arms are stretched horizontally at a distance of* $l = 1\,\mathrm{m}$ *from the centres of mass of their bodies. They are moving along parallel lines, but in opposite directions. The distance between their centres of mass at the moment when they are passing by is exactly* $2l$. *Determine the frequency (in* Hz*) with which they would rotate around their common centre if they grabbed each other's hands. Neglect the mass of their bicycles and energy loss during the act of catching each other. Consider the cyclists to be mass points.* *Mirek making friends with a birch while riding on a bicycle*

Problem FoL.7 ... (non)broken thermometer

We have a particular thermometer with which we measured the temperature of melting ice a $_0 = -0.3\,^\circ\mathrm{C}$ *and the temperature of water vapor as* $t_\mathrm{v} = 101.4\,^\circ\mathrm{C}$*. What is the real boiling poin* *of methanol at atmospheric pressure, if the thermometer shows a temperature* $t = 65.5\,^\circ\mathrm{C}$ *Assume that the real temperature of melting ice is* $\tau_0 = 0\,^\circ\mathrm{C}$ *and of water vapor* $\tau_\mathrm{v} = 100\,^\circ\mathrm{C}$ *Also assume that the number of ticks on the broken thermometer is proportional to the rea emperature in Celsius degrees.* *Lydka played with a thermometer during laboratory practice*

Problem FoL.8 ... line of apsides

Consider a terrestrial planet which orbits its maternal solar-type star in such a way, that w
can neglect other bodies in the universe and the eccentricity the of planet's orbit is $e = 0.150\,0$
Determine the ratio $v_\mathrm{p}/v_\mathrm{a}$ *of the speed of the planet at periapsis* v_p *(the nearest point to th*
star) to the speed at apoapsis v_a *(the farthest point from the star) .* *Karel exercising*

Problem FoL.9 ... the fire rises

What is the lowest temperature of the air (in °C*) inside a balloon, which will make it float*
The volume of the balloon is $V = 6 \cdot 10^6\,\mathrm{l}$ *and its weight (without the air inside) is* $m = 700\,\mathrm{kg}$
The density of surrounding air is $\varrho = 1.2\,\mathrm{kg{\cdot}m^{-3}}$ *and the atmospheric pressure is* $p = 101\,\mathrm{kPa}$

Xellos was baneposting

Problem FoL.10 ... sunny

Mirek decided to do something for his health, so he took his schoolbooks outdoors for fresh air. The sky was clear and the sun was exactly at the south, $\alpha = 50^\circ$ above the horizon. If Mirek places his book horizontally, then it receives radiation power P from the sun. Determine the lowest angle β by which Mirek must tilt the book towards himself (he looks at it from the north) so that the incident radiation power drops to $P/2$. *Mirek's paper was too white*

Problem FoL.11 ... the potassium foil

When Kiki was young and used to not know anything about quantum physics (she only knew the classical view that energy is delivered continuously by a light beam), her home was lit with an isotropic light source with an output of $P = 1.5\,\mathrm{W}$. At a distance $R = 3.5\,\mathrm{m}$, she placed a foil made of potassium with work function $\varphi = 2.2\,\mathrm{eV}$. Help Kiki determine how many seconds it'd take for the foil to absorb enough energy to emit an electron. Assume that the foil absorbs all the incident energy, and that an emitted electron absorbs the energy incident on a circular surface of radius $r = 5 \cdot 10^{-11}\,\mathrm{m}$, given by the typical radius of an atom.

Dominika gazed into the past

Problem FoL.12 ... tuned

A source plays a tone with frequency $f_0 = 440\,\mathrm{Hz}$ (a'). What is the speed of a car movin straight towards the source, if the driver hears b' flat ($f = 466\,\mathrm{Hz}$)? The speed of sound in th air is $v = 343\,\mathrm{m{\cdot}s^{-1}}$, the effects of the material of the car can be neglected.

Meggy wanted a musical problem

Problem FoL.13 ... pitch-black

The retina of the human eye is the most sensitive to the yellow-green colour $\lambda = 550\,\mathrm{nm}$*, fo*
which the sensitivity threshold is $P = 1.7 \cdot 10^{-18}\,\mathrm{W}$*. What is the minimal number of photon*
hat have to hit the retina over one second for the light to be percepted?

Verča had total visual blackou

Problem FoL.14 ... pyramid

A school model of a regular four-sided pyramid was made of wood with mass density $\varrho_1 =$ $= 600\,\mathrm{kg\cdot m^{-3}}$. *The pyramid had mass* $m_1 = 300\,\mathrm{g}$ *and the ratio between the length of its bas edge and the height from the base to the apex was* $2 : 3$. *One day, naughty students brok into physics room and sawed off the poor pyramid's apex. The section was made parallel t the base. The teacher made a new apex with the same size for the pyramid; however, it wa made of wood with mass density* $\varrho_2 = 900\,\mathrm{kg\cdot m^{-3}}$ *and then joined with the truncated pyramic Although the pyramid looks the same now, its mass is* $m_2 = 309.6\,\mathrm{g}$. *Determine the distanc from the base (in* cm*) at which the section was made.* *Meggy remembered solid geometry*

Problem FoL.15 ... battering ram vol. 2

As a part of their offensive, the orcs of Mordor want to break through the gate of Mina Tirith. In a parallel-universe Middle-earth, no progress is made even after calling for Gronc a fire spitting battering ram. Feeling hopeless, Sauron eventually decides to prop up anothe battering ram horizontally between the gate and a newly built perfectly rigid wall. He then orders to heat the ram so as to exploit thermal expansion properties of its material to break the gate. Find out by how much (in Kelvins) the ram needs to be heated. Assume that the ram has the shape of a cylinder with axis perpendicular to the gate. The material of the ram has linear expansion coefficient $\alpha = 1.2 \cdot 10^{-5}\,\mathrm{K}^{-1}$, Young's modulus of compression $E = 211\,\mathrm{GPa}$ and the maximum compressive stress that the gate endures is $\sigma_{\max} = 400\,\mathrm{MPa}$. Consider the thermal expansion process to be fully linear and assume that there is no deformation of the gate until it yields.

Ondra watched El Seňor de los Anillos

Problem FoL.16 ... refracter

A ray of white light is incident on a thin glass wall of a hollow prism filled with carbon disulphide The angle of incidence (measured from the normal to the interface) is $\varepsilon_1 = 50°$. *The walls o the prism are made from thin plane-parallel plates. The apex angle of the prism is* $\varphi = 60°$ *Compute the angular width of the light exiting the prism (in degrees). The refractive index o the glass is* $n_{\mathrm{cs}} = 1.518$ *for red light and* $n_{\mathrm{fs}} = 1.599$ *for violet light; the refractive index o carbon disulphide for red light is* $n_{\mathrm{c}} = 1.618$ *and for violet light* $n_{\mathrm{f}} = 1.699$. *Assume that th surrounding air has refractive index* $n = 1$ *for all wavelengths.*

Faleš struggled with old physics exercise books

Problem FoL.17 ... the Long Night

What amount of work must be done to extend a day by 4 hours? Consider the Earth to b
a homogeneous solid sphere with radius $R = 6\,378\,\mathrm{km}$ and mass $M = 5.972 \cdot 10^{24}\,\mathrm{kg}$.

Kuba would like to have more time to sleep

Problem FoL.18 ... thick glass

Lukáš wants to admire the Black rock from his dormitory window, but another building i standing in the way. Lukáš is $d = 400\,\mathrm{m}$ away from the building and looks (approximately ight at it, so the building appears as a rectangle with height $h = 70\,\mathrm{m}$ and width $w = 50\,\mathrm{m}$ Determine the solid angle (in steradians) covered by the building.

Mirek realized there's landmark next to the dormitory

Problem FoL.19 ... wrong slope

Construction workers don't always follow the blueprints precisely. After one "hard night", they read the plans incorrectly and built a road curve tilted to the other side than it is usually done. By how many kilometers per hour did the maximum speed, with which a car may cross the curve without slipping, decrease? The radius of the curve is $r = 100\,\mathrm{m}$, *the angle by which it's tilted is* $\alpha = 5°$ *and the static friction coefficient between a tyre and the road surface is* $f = 0.55$. *The acceleration due to gravity is* $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$. *Pikoš was watching construction workers*

Problem FoL.20 ... minimal force

Consider three particles with positive charges $Q_1 = 1\,\mathrm{C}$, $Q_2 = 2\,\mathrm{C}$ and $Q_3 = 4\,\mathrm{C}$ placed i
vacuum. Find the minimal possible magnitude of the force between Q_1 and Q_3, if you know
that the force between Q_1 and Q_2 is $F_{12} = 1\,\mathrm{N}$ and the force between Q_2 and Q_3 is $F_{23} = 4\,\mathrm{N}$

Náry was trying to minimise his repulsiveness

Problem FoL.21 ... laser pointer

A monochromatic laser points perpendicularly at a solid screen. We mark the point on th screen at which the laser is pointing when there is nothing but air in between. Now, we plac a transparent plate made from an optically denser material with thickness $d = 2\,\mathrm{cm}$ betwee the laser and the screen. The laser beam enters the plate at an angle $\alpha = 30^\circ$ from the norma and emerges on the opposite side of the plate. The laser is now pointing at a point which i shifted from the original one by $\delta = 0.6\,\mathrm{cm}$. What is the refractive index n_2 of the plate? Th refractive index of air is $n_1 = 1$.

Meggy tried geometry

Problem FoL.22 ... when to stop a catapult

Consider a massless rod in Earth's homogeneous gravity field. The rod rotates about an axi
that is perpendicular both to the gravitational acceleration and to the rod. At a distance R =
= 1 m from the axis, we place a weight $M = 10\,\mathrm{kg}$, and on the other side, we place a projectil
with mass $m = 500\,\mathrm{g}$ at a distance $r = 10\,\mathrm{m}$ from the axis. Afterwards, we set the catapul
(rod) to horizontal position. When the catapult is ready, we release the rod and let it rotat
until it reaches an angle of φ with the horizontal, where it's stopped abruptly. Find the angle φ

hat maximizes the range of the catapult. To simplify the calculations, we assume that th projectile hits the ground at the same height as the height at which it was released when th od was stopped. Also, the projectile cannot slide on the rod.

Lubošek wanted to shoot at his roommate

Problem FoL.23 ... alternating-direct 1

Consider the circuit in the figure 3, the DC source has voltage $U_\mathrm{j} = 4.5\,\mathrm{V}$*, the AC source ha*
voltage amplitude $U_\mathrm{s} = 5\,\mathrm{V}$ *and frequency* $f = 50\,\mathrm{Hz}$*. The resistor's resistance is* $R = 100\,\mathrm{k}\Omega$
and the capacitor's capacity is $C = 10\,\mathrm{nF}$*. What's the expected charge on the capacitor, in* nC

Xellos doesn't like broccoli, that's why he made a problem without it

Problem FoL.24 ... tick tock

There are two clocks in a box. One is a balance wheel clock (like a usual watch; alternatively a digital clock) and the other a pendulum clock – its pendulum consists of a rod (length l = = 30 cm and mass $m = 300\,\mathrm{g}$) with a disk at its lower end (diameter $d = 10\,\mathrm{cm}$ and area density $\sigma = 7.5 \cdot 10^{-2}\,\mathrm{kg{\cdot}m^{-2}}$). The disk is attached lengthwise in the only direction in which the pendulum can move (oscillate). A small rocket lifts the box with acceleration $a = 5g$ (g is the gravitational acceleration of the Earth) up to the height $h = 30\,\mathrm{km}$, where it lets the box move with unchanged inertia in Earth's gravity. Assume that the gravitational field is homogeneous with $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$ and neglect the drag force. By how much will the times on the two clocks differ (in absolute value) at the highest point of the box's trajectory?

Fales remade the relativistic paradox

Problem FoL.25 ... sleepyhead

Exactly at 6:30 in the morning, a solution of a radiopharmaceutical containing radioactive ^{18}F *with volumetric activity* $330\,\mathrm{MBq\cdot ml^{-1}}$ *was prepared for a patient.* 1 ml *of this solution should have been applied at exactly 8:00 on the same day. However, the patient was late and he took his shot at 8:40. What volume of the solution of the radiopharmaceutical (in ml) should the patient get so that the dose would be equivalent to the originally planned one? The half-life of* ^{18}F *is* 109 min *and its decay product is stable oxygen.*

Wake up early and don't miss your deadlines

Problem FoL.26 ... Martian

What is the maximal pressure (in μPa*) that a spherical balloon the size of Mars can hold, if it' made from monolayer graphene? The radius of Mars is* $R_{\mathrm{M}} = 3\,390\,\mathrm{km}$*, the maximal tensio in one layer of graphene is* $\sigma_{\max} = 42\,\mathrm{N{\cdot}m^{-1}}$*. Neglect the gravitation of the gas.* *Elon Musk*

Problem FoL.27 ... nitrogen suffocates the audience

What volume V would the nitrogen from a $V_\mathrm{D} = 7.0\,\mathrm{l}$ Dewar flask take after boiling away i
tandard conditions, i.e. $t = 20\,^\circ\mathrm{C}$ and $p_\mathrm{a} = 1\,013\,\mathrm{hPa}$? We are interested only in the volum
f the evaporated nitrogen, so you can also consider an equivalent problem of determining th
olume of a room which contained only vacuum and liquid nitrogen at the beginning and onl
N_2 gas at standard temperature and pressure after the nitrogen evaporated. The density o
iquid nitrogen is $\varrho_\mathrm{L} = 808\,\mathrm{kg{\cdot}m^{-3}}$, the molar mass of nitrogen is $M_\mathrm{m} = 28.0\,\mathrm{g{\cdot}mol^{-1}}$, th
olar gas constant is $R = 8.31\,\mathrm{J{\cdot}K^{-1}{\cdot}mol^{-1}}$ and the density of nitrogen gas under standar
onditions is $\varrho_\mathrm{G} = 1.16\,\mathrm{kg{\cdot}m^{-3}}$. *Kare*
as wondering what would have to happen to kill the audience of a liquid nitrogen experimen

Problem FoL.28 ... exoplanet

The exoplanet Kepler-138c has radius $R_K = 1.2\,R_E$, where R_E is the radius of the Earth, an density equal to that of the Earth. The inhabitants dug a tunnel straight through the center o the planet (along the diameter). How much longer will it take to free-fall through this tunne (without friction or propulsion) than through a similar tunnel on Earth? Give your answer i seconds. Assume that both planets are homogeneous. *Filip was digging in his garden*

Problem FoL.29 ... a hideout on the Moon

Consider two people (let's call them Iva and Radek) standing on the surface of the Moon. Fin
their smallest possible separation needed so that a voyeur (let's call him Aleš), standing on th
summit of Mt. Palomar in the distance $r = 3.8 \cdot 10^5\,\mathrm{km}$ *from the pair, would be able to resolv*
them from one another using a telescope with aperture diameter $5\,\mathrm{m}$. *Take Iva and Radek to b*
two point-like sources emitting light with wavelength $500\,\mathrm{nm}$. *Dominika making up gossips*

Problem FoL.30 ... melted down thermistor

A thermistor is a semiconductor component, the resistance of which decreases significantly wit emperature according to the formula $R = R_0 \exp\left(c\,(1/T - 1/T_0)\right)$*, where the temperatures* T T_0 *are substituted in* K, $R_0 = 120\,\Omega$ *is its resistance at room temperature* $T_0 = 25\,^\circ\mathrm{C}$ *an* $= 3.0 \cdot 10^3\,\mathrm{K}$*. This formula loses validity at the temperatur* $T_\mathrm{t} = 150\,^\circ\mathrm{C}$*, when the thermisto tarts to melt down. The power output of heat transfer from the thermistor at a temperatur* T *to its surroundings at the temperature* T_0 *is* $P_\mathrm{chl} = k(T - T_0)$*, where* $k = 4.5 \cdot 10^{-4}\,\mathrm{W{\cdot}K^{-1}}$ *Determine the highest voltage to which the thermistor can be connected without being heate up to the temperature* T_t *or higher (after a sufficiently long time).*

Xellos remembered his laboratory practice

Problem FoL.31 ... rotation rotation

Verča had a fancy party hat on, but she moved too fast and it fell to the ground and began t roll. The conical hat has a slant height of $R = 30\,\mathrm{cm}$, the radius of the base circle is $r = 10\,\mathrm{cm}$ and during its movement, the cone rotates about its stationary apex with an angular velocity ω_y The cone also rotates about its rotational axis with an angular velocity $\omega_o = 5.0\,\mathrm{rad{\cdot}s^{-1}}$. What is the angular velocity of any point on the surface of the cone with respect to the immediat axis of rotation? The cone rolls on a rough surface. *Mirek combined rotations*

Problem FoL.32 ... superpowers

f Aleš were to pick a superpower, he would choose the ability to see the intensities of electri
nd magnetic field. Let's assume that he has such an ability. Imagine that he stands 1.8 m *awa*
rom a point source of light with power $P = 250\,\mathrm{W}$. *What is the effective value of the magneti*
ield (magnetic induction) which he feels?

Dominika thinks of the consequences of the organizers' wishes

Problem FoL.33 ... ball-ey

A small homogeneous spherical ball with radius $r = 3\,\mathrm{cm}$ is placed in a fixed spherical hole wit
radius $R = 10\,\mathrm{cm}$. The ball can roll without slipping. We move the ball from its equilibriu
position a little bit. What will be the period of its oscillations?

Guljočka v jamočke, Tom kept repeating

Problem FoL.34 ... diodes

Consider two diodes satisfying the ideal Shockley diode law

$$I(U) = I_{\mathrm{S}} \left(\exp\left(\frac{U}{V_{\mathrm{T}}} \right) - 1 \right) ,$$

connected in series with a DC source with voltage $V = 1\,\mathrm{V}$ *and internal resistance* $100\,\Omega$*. Th*
saturation current of one diode is $I_{\mathrm{S}} = 1 \cdot 10^{-11}\,\mathrm{A}$*, the saturation current of the other one i*
twice as large. The thermal voltage is $V_{\mathrm{T}} = 26\,\mathrm{mV}$*. What is the total electric power (in* mW
produced on the diodes?

Hint *Since we only want a numerical result, it's sufficient to solve the equations numerically*

Jančí was thinking about diodes in series

Problem FoL.35 ... gardener's problem

An amateur gardener is watering his garden. He took a cylindrical bucket with base area $S =$ $= 420\,\mathrm{cm}^2$ *and filled it with water up to a height* $h = 40\,\mathrm{cm}$. *Afterwards, he made a hole wit* *rea* $s = 2\,\mathrm{cm}^2$ *in the bottom base and started watering. What's the longest row of plants whic* *e's able to water, if his speed is* $u = 1\,\mathrm{m\cdot s}^{-1}$?

Hint *The speed with which water flows out of the hole is given by the formula* $v = \mu s\sqrt{2g}$ *with a coefficient* $\mu = 0.6$. *where* x *is the current water level height.*

Marek was watering his garden

Problem FoL.36 ... a merry encounter

In a vast empty space, there are six electrons very far from each other, located in the vertices o a regular hexagon. Each of them is moving towards the center of this hexagon with the speed $v = 1\,000\,\mathrm{m\cdot s^{-1}}$. How close to each other will two adjacent electrons get before the repulsiv force separates them forever?

Mirek was inspired by Náry's struggles

Problem FoL.37 ... Achilles and the tortoise

Consider an optical fibre which consists of a core, with diameter $d = 50\,\mu\mathrm{m}$ *and refractiv*
ndex $n_1 = 1.460$, *which is surrounded by cladding with refractive index* $n_2 < n_1$. *The ray*
propagate inside the core due to total internal reflection. A measurement was performed o
fibre with length $l = 500\,\mathrm{m}$ *and it was estimated that the time dispersion (which is due t*
he rays propagating along different paths) at the receiving end was $\Delta t = 20\,\mathrm{ns}$. *Determin*
he refractive index n_2 *of the cladding. Assume that the speed of light in vacuum is* $c =$
$= 2.997\,9 \cdot 10^8\,\mathrm{m{\cdot}s^{-1}}$ *and that light is not dispersed inside the fibre.* *Michaleus*

Problem FoL.38 ... this problem sucks

Find the radius (in millimeters) of the largest lead ball which we are able to suck up usin a vacuum cleaner with suction power 200 W. *The machine is equipped with a plain hose wit a circular cross-section of diameter* 6 cm. *To simplify matters, assume that the ball is alread hovering in the air high above the floor. Take the densities of air and lead to be* $\varrho = 1.2\,\mathrm{kg{\cdot}m^{-}}$ *and* $\varrho_{\mathrm{Pb}} = 11\,340\,\mathrm{kg{\cdot}m^{-3}}$, *respectively.* *Kuba lost his pen-drive in a peculiar way*

Problem FoL.39 ... bursting with rage

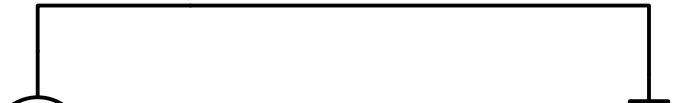
Kiki found a hair band and didn't know what to do with it, so she put it around Náry's head
The length of the unstretched band is $l_0 = 15\,\mathrm{cm}$*, its length when put around Náry's head i*
$= 55\,\mathrm{cm}$*. The cross-section of the band has a square shape with a constant side length o*
$= 2\,\mathrm{mm}$*. Let us naively assume that the band is behaving linearly according to Hooke's law*
and has Young's modulus $E = 50\,\mathrm{MPa}$*. What is the pressure (in* kPa*) crushing Náry's head*
Take the head to be perfectly spherical and the band to be lying on its great circle.

Watermelon gore by Mirek

Problem FoL.40 ... alternating-direct 2

Consider the circuit in the figure, the DC source has voltage $U_\mathrm{j} = 4.5\,\mathrm{V}$*, the AC source ha*
voltage amplitude $U_\mathrm{s} = 5\,\mathrm{V}$ *and frequency* $f = 50\,\mathrm{Hz}$*. The resistor's resistance is* $R = 100\,\mathrm{k}\Omega$
and the capacitor's capacity is $C = 10\,\mathrm{nF}$*. What's the maximum charge on the capacitor, in* nC

Xellos still doesn't like broccol



Problem FoL.41 ... ambulance

Verča wanted to pretend being an ambulance. She took a honking siren and started runnin from Karel towards the door at a speed $v = 7\,\mathrm{km{\cdot}h^{-1}}$. *Karel, who watches the situation wit interest, hears interference beats at a frequency* $f = 6\,\mathrm{Hz}$. *What is the frequency* f_0 *of the siren The speed of sound in the air is* $v_\mathrm{s} = 340\,\mathrm{m{\cdot}s^{-1}}$.

Verča is starting to go crazy from all that physics

Problem FoL.42 ... transmittors

Imagine that we have two plane-parallel mirrors and we know that each mirror transmits onl
a fraction of incident light. The first mirror transmits $t_1 = 2/3$ of incident light, the second onl
$t_2 = 1/3$. What fraction T of incident light passes through our system of two parallel mirrors
No energy is lost in the mirrors.

Karel pondered what "showing someone a mirror" meant in the former Communist Bloc

Problem FoL.43 ... homemade maglev

Mišo adores trains and desperately wants to take a ride on a maglev - a train that uses magneti levitation to move without touching the ground. In order to avoid travelling far abroad, h decided to build his own maglev at home. He plans to use a set of electromagnets in the shap of the letter U *to create the required magnetic field. The inductor is wound on a core and ha* $N = 100$ *turns. The cross section of the core is* $S = 5\,\mathrm{cm}^2$ *and it's constant along its whol length. The length of the central line of force inside the inductor is* $l = 20\,\mathrm{cm}$*. The core ha relative permeability* $\mu_\mathrm{r} = 1\,000$*. To test the strength of the magnets, Mišo places a steel girde next to both ends of one magnet, then connects the inductor to a current source and tries to pu the girder away. How large will be the current passing through the inductor if he needs a forc of* $F = 100\,\mathrm{N}$ *to pull the girder away? Consider the permeability of air to be approximately th same as the permeability of vacuum* $\mu_0 = 4\pi \cdot 10^{-7}\,\mathrm{H{\cdot}m^{-1}}$*. Neglect the weight of the girde*

Mirek tried to find a "practical" use for excercise problems

Problem FoL.44 ... accretion is too mainstream

According to some theories, there used to be a 10th planet of the Solar System called Phaeto
between Mars and Jupiter. Its distance from the Sun is supposed to have been approximatel
$D_P = 2.5\,\mathrm{au}$ *and its radius* $r_P = 1\,000\,\mathrm{km}$. *Determine the surface temperature of Phaeton afte*
radiation equilibrium is established, if its Bond albedo is similar to that of Earth, $A = 0.$
and so is emissivity, which is approximately 1). Phaeton is supposed to have existed at a tim
when the Sun had surface temperature $T_S = 5\,000\,\mathrm{K}$ *and radius* $r_S = 6 \cdot 10^5\,\mathrm{km}$.

Mirek has been thinking about moving somewhere else

Problem FoL.45 ... wavy electron

What wavelength will an electron have, if we speed it up from rest with electric field ove
potential difference of 60 kV*?* *Kiki and her introduction to quantum physics for dogs*

Problem FoL.46 ... rising bubble

Having had a great time doing our laundry in a sink, we managed to release a number of soa bubbles into the air. Some of them have burst after flying around for a short while, but a numbe of them happened to ascend relatively high up – the most successful guy reached the altitude o $H = 1\,500\,\mathrm{m}$ *above ground level. How many times did the volume of this bubble increase durin its ascent? Use the ideal gas law and the equation of hydrostatic equilibrium applied to th atmosphere in order to find the pressure as a function of altitude above ground level. Assum that there is a linear relation of the form* $T(h) = T(0) - kh$ *between the temperature and th altitude, where* $k = 0.006\,5\,\mathrm{K{\cdot}m^{-1}}$*. The atmospheric pressure at ground level is* $p(0) = 101\,\mathrm{kPa}$ *the temperature at ground level is* $T(0) = 25\,^{\circ}\mathrm{C}$*. The molar mass of air is* $M = 29.0\,\mathrm{g{\cdot}mol^{-1}}$ *Assume that the pressure inside and outside the bubble is the same.*

Mirek doing calculations and laundry at the same time

Problem FoL.47 ... a charged spiral

Consider a charged non-conducting spiral parametrised in 2D cartesian coordinates as

$$\{[l_0 t \sin \ln t, l_0 t \cos \ln t], t > 0\} .$$

Assume that the linear charge density ϱ along the spiral can be expressed as a function of t namely $\varrho(t) = \varrho_0 t \exp(-t^2)$. Find the value of electric potential at the origin and express it a multiple of ϱ_0/ε_0. The spiral is placed in vacuum. *Due to Jančí*

Problem FoL.48 ... washing the dishes

Verča likes to wash the dishes. When she's done, she takes a little boat and places it on th surface of water mixed with detergent. The boat is a small wooden board with thickness b = = 0.5 cm, length $l = 10\,\mathrm{cm}$ and width $w = 5\,\mathrm{cm}$. Its length is parallel to the axis x. The surfac tension in the x-direction is given by $\sigma(x) = \sigma_0 + xs$ and in the y-direction by $\sigma(y) = \sigma_0$ where $\sigma_0 = \mathrm{const}$ and s is the gradient of surface tension. Determine the intial acceleration o boat after being placed on the water surface. The density of wood is $\varrho = 800\,\mathrm{kg{\cdot}m^{-3}}$, σ_0 = = $30\,\mathrm{mN{\cdot}m^{-1}}$, $s = 80\,\mathrm{mN{\cdot}m^{-2}}$. If the acceleration is in the direction of increasing surfac tension, it's positive; if it's in the opposite direction, it's negative. The contact angle of woo and water is $\beta = 45°$. The inclination of the boat is negligible.

Mirek watched students washing the dishes during the summer camp

Problem FoL.49 ... amplifying

Find the gain of the circuit in the figure. The gain is defined as $g = U_{\text{output}}/U_{\text{input}}$ *(includin* *he sign, the voltages are calculated with respect to the ground). All resistors in the circui* *ave resistivity* $R = 10\,\text{k}\Omega$*. Assume that the op-amp is ideal.*

Hint *To solve this problem, you will only need Ohm's law, Kirchhoff's first law and th* *ollowing information: The triangle symbol in the circuit is an op-amp (operational ampli* *ier). It has two inputs (− and +) and one output (the third vertex of the triangle), and follow* *hese rules:*

- *there is no current going in or out of any of the inputs,*
- *the op-amp sets the voltage at its output so that the voltage difference between its input* *is zero.*

Pikoš likes playing with complicated circuits

Problem FoL.50 ... watering

In a pool full of water, there is a vertical tube with internal cross-section $S_1 = 4\,\mathrm{cm}^2$*. On it top, it opens into a cylindrical container with a few small holes on the perimeter. The whol construction rotates and the cylindrical container is radially partitioned, so that the wate rotates with the container. The cylinder has radius* $r = 10\,\mathrm{cm}$ *and the total cross-section of th holes is* $S_2 = 5\,\mathrm{mm}^2$*. At first, the holes are closed and the whole tube and cylinder are fille with water and rotated with an angular speed* $\omega = 25\,\mathrm{rad{\cdot}s^{-1}}$*. Then, the holes are opened so that water can flow out. Determine the speed* v *of the water flowing out of the holes wit respect to the laboratory reference frame (which doesn't rotate with the container). The ai pressure is* $p_0 = 1\,015\,\mathrm{hPa}$*, the acceleration due to gravity is* $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$ *and the hole in the cyliner are* $h = 20\,\mathrm{cm}$ *above the surface of the pool. Consider the density of water t*

be $\varrho = 1\,000\,\mathrm{kg \cdot m^{-3}}$.

Faleš was inspired by the Estonian-Finnish olympiad, which had a partially wrong solution

Problem M.1 ... submarine

Michal is planning a journey from water level at the Mariana Trench ($0\,\mathrm{m}$ *above sea level) dow*
to the seabed ($10\,971\,\mathrm{m}$ *below sea level). What difference in hydrostatic pressure (in* MPa*) wi*
Michal's submarine measure? Assume that water density doesn't change with depth. Conside
the constants $\varrho_{\mathrm{H_2O}} = 1\,000\,\mathrm{kg{\cdot}m^{-3}}$ *and* $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$.

Zuzka was thinking about a submarine

Problem M.2 ... wet FYKOS-bird

The FYKOS-bird is riding on a tricycle with speed $v = 10\,\mathrm{km\cdot h^{-1}}$*, when, to his displeasure t starts raining. Not only is he getting wet, but he also has to increase his power output i order to keep moving at the same speed. Assume that the surface exposed to the uniform rai s* $S = 0.5\,\mathrm{m}^2$ *and that on average, 2 droplets fall on each* cm^2 *over time* $\Delta t = 1\,\mathrm{s}$*. One drople veighs* $m = 0.1\,\mathrm{g}$*. The rain falls straight down, the bird is moving in a straight horizontal line a onstant speed. Neglect the fact that the FYKOS-bird has to move with respect to the tricycle By how much does he need to increase his output power?*

Faleš got wet

Problem M.3 ... water pump

A tube of diameter $r = 2\,\mathrm{cm}$, bent to the shape of the letter L*, has one end submerged i a water tank. Its lower part is parallel to the water surface and the other part sticks out fro the water vertically. The tube moves in the direction of the line connecting the bend and th submerged end, with speed $v = 2\,\mathrm{m{\cdot}s^{-1}}$. How high above the water surface (in* cm*) in the tan will the water in the tube rise? Consider the acceleration due to gravity to be $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$*

Faleš saw a nice picture of a bent tube

Problem M.4 ... rocket bird

The Fykos-bird's wings hurt, so he decided to use physics and build a rocket-propelled backpack
During ignition, the backpack consumes 100 g *of fuel per* 1 s *and exerts a* 50 N *propulsion force*
What's the speed of exhaust gases with respect to the backpack? The bird with the jet-pack i
much heavier than the consumed fuel. *Faleš's legs hurt*

Problem E.1 ... trinity

The resistance R_x is equal to the resistance of the whole circuit. The resistances R_1 and R_2 ar
he same. Compute the value of R_x as a multiple of $R = R_1 = R_2$.

Faleš liked the golden ratio

Problem E.2 ... Náry brawls with numbers

An electromagnetic plane wave is travelling through vacuum. We have measured the value of electric intensity $\boldsymbol{E}$ *and magnetic induction* $\boldsymbol{B}$ *at a particular place in space. The resulting values are* $\boldsymbol{E} = (2, 3, 1)\,\mathrm{V{\cdot}m^{-1}}$, $\boldsymbol{B} = (5, -3, -1)\,\mathrm{T}$. *We are interested in the direction in which the wave is travelling. Compute this direction, normalize the direction vector (multiply it by a positive number such that the resulting vector has unit length) and send us sum of its three components.* *Náry was rotating while being normalized by field theory knowledge*

Problem E.3 ... Náry works hard

When he was supposed to move a telescope, Náry rather moved a charge $Q = 2\,\mathrm{C}$ from infinit
o a place with electric potential $\varphi = 2\,\mathrm{kV}$. How much work (in mJ) did he do? Consider th
potential at infinity to be zero. *Náry and Faleš at the theoretical physics camp*

Problem E.4 ... capacitOr

A dielectric medium between two circular plates of a capacitor has two layers. The first on
is air, with thickness $d_1 = 2\,\mathrm{mm}$, *and the second is acrylic glass with thickness* $d_2 = 4\,\mu\mathrm{m}$
Determine the capacity of the capacitor (in pF*), if the area of each plate is* $S = 2\,\mathrm{dm}^2$. *Th*
permittivity of vacuum is $\varepsilon_0 = 8.854 \cdot 10^{-12}\,\mathrm{C}^2{\cdot}\mathrm{N}^{-1}{\cdot}\mathrm{m}^{-2}$.

Faleš couldn't read a number in his old notebook, so he changed the dimensions

Problem X.1 ... we're expanding

What amount of work (in kJ*) is done by an ideal gas during isothermal expansion, if its initia*
volume is $V_1 = 10\,\mathrm{dm}^3$ *and initial pressure is* $p_1 = 1.0\,\mathrm{MPa}$*? The final pressure of the gas i*
$p_2 = 100\,\mathrm{kPa}$. *Marek doesn't remember where he got it from*

Problem X.2 ... fiat lux

Every time Domča gets all sad and weary in the evening, all she needs to do is switch her lam
on and the incoming photons always cheer her up. What is the number of photons emitte
by a 100-watt lamp over a period of one second? Assume that the lamp emits yellow ligh
with a wavelength of 580 nm *with* 10 % *efficiency and the photons with other wavelengths ar*
considered losses. *Kiki hates transitions to daylight saving*

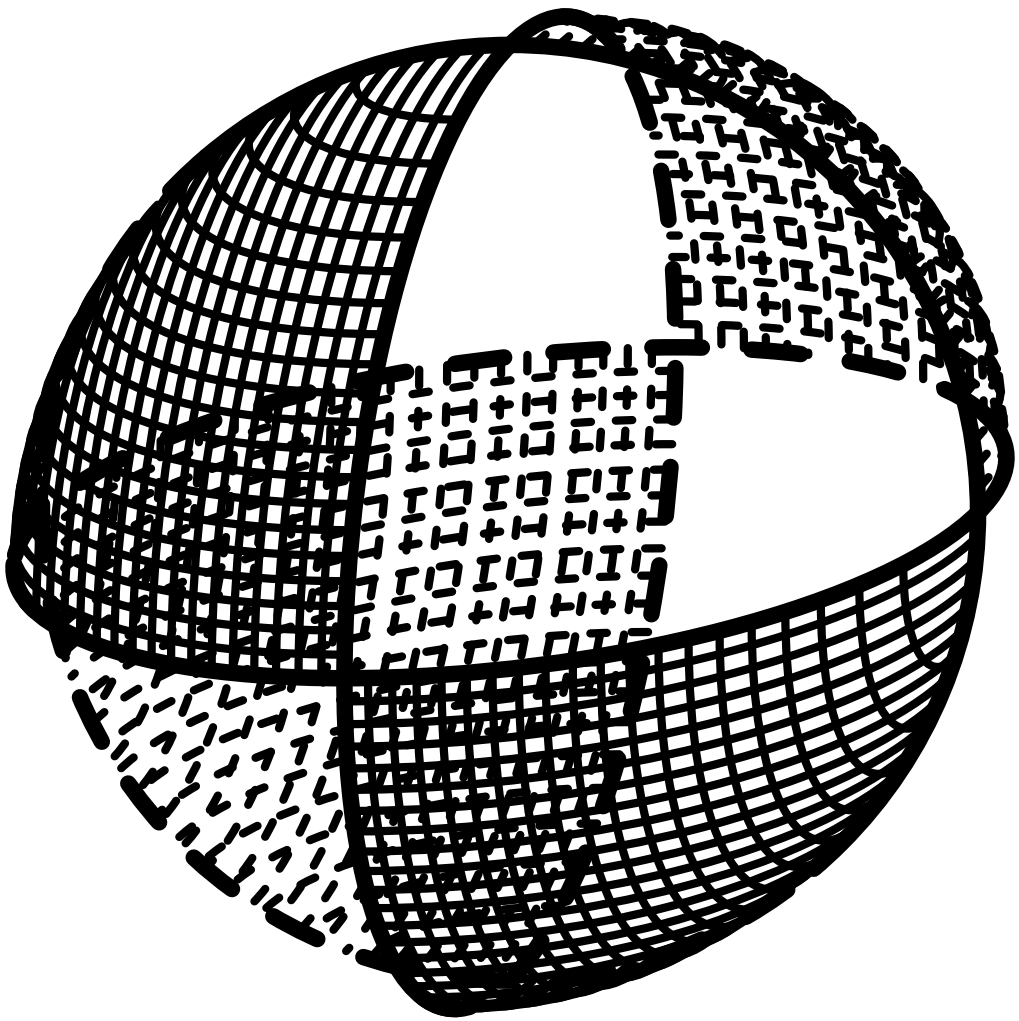
Problem X.3 ... Christmas

For a three-dimensional object, a line o is called an n-fold axis of symmetry if the object doe
not change upon rotation around this axis by an angle $2\pi/n$. For example, a regular heptago
(regular planar polygon with 7 edges) has one 7-fold axis of symmetry, perpendicular to th
heptagon and passing through its center.

Jančі was decorating cookies for Christmas. One of them was shaped like a sphere. Janč
divided the surface into eight equal parts and coloured them alternately, such that the coloure

parts of the surface touched only in vertices. How many 3-fold axes of symmetry does thi
cookie have?

Jančі was helping in the kitchen



Problem X.4 ... antichemical

One of the rules governing the filling of electron orbitals in atoms is Pauli's exclusion principle which says that two electrons can never be in the same state. This gives us the usual rule o two electrons in one orbital (with the same quantum numbers n, l and m), because they ca differ by the orientations of their spins (this is the fourth quantum number m_s).

Imagine that we changed this rule and now, two electrons can be in the same state, but no three. How many valence *electrons would a neutral sulphur atom have, if the orbitals were sti being filled in the same order as for normal electrons. Lada was thinking about fermions*